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Multidimensional decomposition framework for electrocardiographic interference, noise and artifacts removal from diaphragmatic electromyography

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ABSTRACT

Background and Objective: Accurate extraction of respiratory diaphragmatic electromyography (dEMG) is hindered by electrocardiographic (ECG) interference, motion artifacts, and noise. Existing methods, although effective, may remove physiologically relevant information. To enable robust ECG and artifact suppression while maintaining respiratory-related amplitude variations, a higher-order singular value decomposition (HO-SVD) based approach was developed.

Methods: The HO-SVD algorithm exploits the multidimensional structure of dEMG recordings by jointly decomposing the temporal, overlapped QRS complexes, and spatial (channel) dimensions. The method enables targeted suppression of coherent and repetitive artifacts prior to signal reconstruction. Performance was evaluated on three synthetic datasets and on clinical data from ten patients with obstructive sleep apnea, using a previously developed two-dimensional SVD algorithm (SVD_{RI}) as a reference.

Results: Across synthetic datasets, HO-SVD consistently outperformed SVD_{RI} in all evaluation metrics. In clean conditions, HO-SVD improved the median signal-to-noise ratio (SNR) by 27 dB (131%) relative to the original signal. In datasets with mild motion artifacts and complex muscle interference, median SNR improvements of up to 19 dB (88%) were observed. The symmetric mean absolute error and mean frequency deviation were significantly lower for HO-SVD across all input SNRs ($p < 0.0001$) compared to SVD_{RI}. In clinical data, HO-SVD achieved a significantly higher correlation with reference esophageal pressure ($p < 0.02$), greater ECG suppression ($p < 0.0001$), and fewer outliers ($p < 0.0001$) than SVD_{RI}.

Conclusion: These results demonstrate that incorporating a third (spatial) dimension and adaptive thresholding across tensor modes enables robust removal of ECG interference and motion artifacts while preserving respiratory dEMG modulation. The proposed HO-SVD framework offers a flexible and effective approach for multichannel biomedical signal denoising.

1. Introduction

Surface electromyography (EMG) is a non-intrusive technique that uses electrodes on the skin surface to record the electrical activity of the muscles. This method is widely used in many fields, including rehabilitation, patient monitoring, and clinical diagnostics. In particular, its potential is growing in the context of respiration, where capturing diaphragm activity provides direct information about one of the key parameters - respiratory effort [1]. Currently, the gold standard metric for estimating respiratory effort is esophageal pressure (Pes), but its measurement is obtrusive and uncomfortable for patients [2]. Although non-intrusive techniques such as respiratory inductance plethysmography or impedance-based methods have been proposed and used, their indirect nature limits their accuracy in reflecting true respiratory effort. Therefore, diaphragmatic EMG (dEMG) has emerged as a promising non-intrusive alternative which, by measuring the diaphragmatic muscle activity more directly, reflects respiratory effort. However, clinical uptake of dEMG to monitor respiratory effort remains strongly limited by signal quality, as recordings are highly contaminated by electrocardiographic (ECG) activity, noise, and motion artifacts. These disturbances often exceed the amplitude of the dEMG itself, effectively

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masking the physiological information of interest and reducing the reliability of the measurement [3,4].

Several approaches have been proposed to mitigate ECG interference in dEMG recordings. Among the most widely used in clinical practice are the gating technique (GT) and high-pass filtering (HPF). GT suppresses the most prominent ECG components by setting short windows around the R-peaks (100 ms) to zero values [5,6]. In HPF, the cut-off frequency is selected based on spectral differences between ECG and EMG signals: while most ECG energy is concentrated below 35 Hz, EMG activity extends into higher frequencies (18–450 Hz) [7]. Consequently, a cut-off of around 30 Hz is typically considered appropriate [8,9]. However, both GT and HPF influence the EMG itself, either by removing signal portions around the R-peaks or by attenuating physiologically relevant low-frequency components, which limits their suitability for accurate dEMG analysis [10].

Beyond these methods, wavelet decomposition has also been explored in the literature [11,12]. The wavelet transform enables the simultaneous decomposition of EMG signals in both the time and frequency domains, providing better discrimination of transient ECG components from the broader band EMG activity. Strategies based on this principle have shown promise for the suppression of ECG components in EMG, but their performance depends heavily on the choice of mother wavelet [12] and on the relative amplitude of ECG and EMG signals, which may vary between subjects and recording conditions, limiting their robustness in practice.

Among the proposed strategies, blind source separation (BSS) methods have been widely investigated for ECG components and artifact removal from EMG recordings. In general, BSS methods are based on the decomposition of mixed signals into underlying sources, with the goal of separating physiological EMG activity from contaminating components such as cardiac signals or motion artifacts [13–15]. Techniques such as canonical correlation analysis (CCA) [16,17], principal component analysis (PCA), independent component analysis (ICA), and independent vector analysis (IVA) [18] have been applied to suppress different types of contamination in EMG, among these PCA and ICA being specifically used to remove ECG components from the EMG signal [14,15,19–22]. PCA identifies orthogonal components based on the covariance structure and can isolate dominant noise contributions, but prioritizes variance rather than physiological relevance and therefore does not guarantee a reliable separation of ECG interference from the EMG signal. ICA extends this framework by maximizing the statistical independence between components, often achieving improved separation of cardiac and muscular sources. However, its performance strongly depends on the validity of the independence assumptions, which are not always met in practice.

Building on these methods, singular value decomposition (SVD) has been introduced as a related factorization technique that offers efficient computation and has shown potential for ECG components removal in EMG [23–25]. In our previous work, we compared the performance of two SVD variants on datasets that included motion artifacts: a calibration-curve SVD [26] and an SVD guided by a frequency ratio index (RI) [27]. The RI method, previously explored in ultrasound and electrohysterography [28–30], significantly outperformed the calibration-curve approach, as well as classical baselines such as gating, high-pass filtering, template subtraction, and ICA for ECG interference and artifact removal from EMG [26,27]. These findings established SVD_{RI} as the strongest-performing method within the family of SVD-based approaches developed for respiratory dEMG decomposition. Although effective, these implementations have not yet taken advantage of the additional information available in multichannel recordings, and the use of a two-dimensional decomposition may in some cases suppress more than the unwanted components, potentially leading to a loss of useful EMG detail.

Recently, higher-order singular value decomposition (HO-SVD) has been introduced in biomedical signal processing to improve the separation of signal components in temporal, spatial, and other dimensions. It has shown enhanced performance over traditional SVD in applications such as Doppler ultrasound imaging for clutter suppression of minimally moving targets [31,32], coronary flow [33], as well as noise reduction in MRI and EEG [34,35]. Motivated by these successful applications, we explored HO-SVD for decomposing dEMG signals, where incorporating the channel dimension may further enhance the suppression of ECG components, motion artifacts, and noise - an approach never reported in this context.

The aim of this study is to develop a method for removing ECG contamination and other sources of interference from dEMG, including mild movement artifacts and high-intensity motion- and muscle-related interference. The proposed algorithm builds on HO-SVD and introduces selection criteria across tensor dimensions to identify singular values representing the unwanted parts in the signal of interest. In this higher-dimensionality framework, the previously introduced ratio index (RI) [27] is complemented by two additional conditions. Our earlier method (SVD_{RI}), shown to outperform classical techniques, was used for comparison to benchmark the new approach on synthetic and clinical datasets. By improving interference removal and interpretability, HO-SVD approach could facilitate a more reliable extraction of respiratory effort from dEMG signals, supporting the broader clinical and research use of this technique.

2. Methods

2.1. Datasets

2.1.1. Synthetic dEMG

Three synthetic datasets were created to simulate respiratory dEMG by combining clean EMG acquired from the skeletal muscles with ECG signals. The ECG data included recordings acquired during sleep (A) without artifacts, (B) during sleep with mild motion artifacts caused by body movements, and (C) during cycling and running with complex motion- and muscle-related interference. This design provided a ground-truth EMG reference to objectively evaluate the proposed method under varying levels of artifact contamination.

Clean EMG recordings were collected from 32 healthy volunteers using high-density EMG grids (8 × 8 channels, 4 mm interelectrode spacing) placed over the biceps and triceps of the dominant arm. The signals were acquired with a Refa amplifier (TMS International, Enschede, The Netherlands) at a sampling rate of 2048 Hz. During acquisition, the participants sat comfortably and performed an isometric elbow flexion at 50% of their maximal voluntary contraction for 60 s. From the available recordings, 20 EMG signals were selected after visual inspection based on signal quality and the absence of visible artifacts. The study protocol was approved by the Ethical Committee of the Máxima Medical Center (Veldhoven, the Netherlands). For each trial, bipolar derivations were obtained from the high-density EMG grid by averaging two groups of four neighboring electrodes aligned with the muscle-fiber direction and subsequently subtracting the two averaged signals. Repeating this procedure across the grid resulted in six bipolar channels. These signals were downsampled to 512/250 Hz to match the ECG sampling frequency. Finally, the EMG segments containing muscle contraction were windowed and concatenated so that their envelope followed a respiratory-like pattern of 14 cycles per minute [36].

ECG recordings along with patient position data were obtained from overnight polysomnography of the SOMNIA dataset, which includes patients with a variety of sleep disorders [37]. Ethical approval was obtained from the medical ethical committee of the Máxima Medical Center (Veldhoven, The Netherlands; File no: N16.074), and all participants provided written informed consent. The study followed the principles of the Declaration of Helsinki. Permission to access the anonymized recordings was granted on July 7, 2022. The ECG signals were recorded using Ag/AgCl electrodes (H34SG) in a lead II torso configuration. Body position was captured using a gravity-detecting

Table 1
Synthetic dEMG datasets details.

Dataset	Signal combinations	Subjects × channels
A	EMG + ECG	20 × 6
B	EMG + ECG with mild motion artifacts	20 × 6
C	EMG + ECG with complex motion and muscle artifacts	19 × 6

electric sensor (Grael position sensor) placed on the front of the thoracic respiratory inductance plethysmography belt. For this work, two sets of 20 ECG segments of 60 s were extracted: one consisting of clean ECG selected from periods without detected position changes (A), and another containing ECG recorded during body movements as identified by position sensors, thus representing signals contaminated by mild motion artifacts (B). These ECG signals were sampled at 512 Hz.

In addition to these two datasets, a third set of ECG recordings was recorded from 11 healthy volunteers (C). All participants signed a written consent. Measurements were performed using Ag/AgCl electrodes and a P-Cardio portable ECG recording device (P-Cardio Wearables, Valkenswaard, The Netherlands). The dataset contained five-minute recordings of sitting, conversing, walking, cycling on an indoor bicycle, and running. For the purpose of this study, 19 recordings during running and cycling were selected to provide an additional level of noise in the ECG signal caused by complex motion and skeletal muscle artifacts. The acquired signals were sampled at 250 Hz.

The EMG and ECG signals were pre-processed with a second-order band-pass filter, set to 5 to 450 Hz for EMG and 0.8 to 120 Hz for ECG [38,39]. To further reduce interference, a 50 Hz notch filter was applied to suppress power line noise. To reproduce realistic acquisition conditions, the EMG recordings were rescaled to match signal-to-noise ratios typically observed in dEMG (ranging from −26 to −16 dB). The rescaled EMG was subsequently superimposed on the ECG traces, creating three datasets: (dataset A) representing clean conditions, (dataset B) including mild motion-related disturbances, and (dataset C) containing interference from high-intensity activity as shown in Table 1.

2.1.2. Clinical dEMG

The real data was provided from the aforementioned SOMNIA [37] dataset. From the complete dataset of 66 patients with obstructive sleep apnea (OSA) who underwent full-night polysomnography, 10 were selected considering the availability of the reference signal (Pes) providing additional evaluation of two methods. The analysis in this paper focused mainly on the dEMG signals and Pes as a reference. The Pes signal was acquired through a flexible catheter equipped with a pressure sensor at its tip positioned transnasally in the esophagus above the diaphragm (Gaeltec CTO-1, sampling frequency = 128 Hz). Four monopolar Ag/AgCl electrodes (H34SG) were used to record dEMG signals placed in the sixth intercostal space (two frontal, two dorsal) with a reference electrode on the sternum, sampled at 512 Hz.

The Pes signal was upsampled at 512 Hz using band-limited interpolation (sinc-type), by a polyphase FIR filter to align with the dEMG sampling rate. Subsequently, a fourth-order band-pass Butterworth filter was applied within the respiratory frequency range of 0.05 to 0.6 Hz to derive the Pes envelope.

dEMG was preprocessed using a fourth-order Butterworth filter with a band-pass range of 5 to 250 Hz [38], along with a 50-Hz notch filter to suppress the power line interference. The bipolar configuration was obtained by subtracting signals from spatially adjacent electrodes: left-frontal with left-dorsal, right-frontal with right-dorsal (front–back pairs), and the two frontal electrodes as well as the two dorsal electrodes (left–right pairs), resulting in four bipolar channels.

2.2. Higher order SVD method

HO-SVD has been proven to achieve promising results in image clutter removal and filtering methods [31–35]. In case of dEMG using SVD in the higher-dimensional tensor allows adding an additional dimension utilizing the multichannel nature of the recordings. That provides the opportunity to select the individual thresholds in higher-order dimensions. In this study, we apply the HO-SVD to decompose a 3D-tensor $T \in \mathbb{C}^{M\times N\times P}$, which is unfolded along each mode obtaining matrices $T_1 \in \mathbb{C}^{M\times NP}$, $T_2 \in \mathbb{C}^{N\times MP}$, $T_3 \in \mathbb{C}^{P\times MN}$. The unfolded matrices are then subjected to the SVD

$$T_k = U_k S_k V_k^H, k = 1, 2, 3, \tag{1}$$

where U_k and V_k contain the left and right singular vectors of the mode-k unfolding, respectively, and S_k is the diagonal matrix containing the corresponding singular values. Here, k denotes the tensor mode. Next, the 3-D tensor T is decomposed into a core tensor $S \in \mathbb{C}^{M\times N\times P}$ and a set of factor matrices $U^{(1)}$, $U^{(2)}$, $U^{(3)}$, which correspond to each mode of the tensors, given as [40]

$$T = S \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \tag{2}$$

where $\times_n$ denotes the outer product of mode-n. For tensor decomposition using HO-SVD the Tensorlab 3.0 MATLAB package was used [41].

The method introduced in this study (HO-SVD) extends a previously developed algorithm for ECG suppression from EMG signals [26]. As in the earlier approach, R-peak positions were first identified using the detector described by Varanini et al. [42]. For each detected peak, a third order tensor $T \in \mathbb{C}^{N_{RR}\times N_{QRS}\times N_{CH}}$ was constructed using samples within each R–R interval along one dimension (N_{RR}), stacking 50 subsequent QRS complexes along the second dimension (N_{QRS}), and adding all channels along the third dimension (N_{CH}). The tensor-construction procedure is illustrated in Fig. 1.

Each slice of T contained a QRS segment starting 0.3 s before the corresponding R peak. The tensor was then decomposed by HO-SVD (Eq. (2)), where $U^{(1)}$, $U^{(2)}$ and $U^{(3)}$ corresponded with the orthogonal singular vector matrices in the temporal, QRS, and channel dimensions, respectively. This higher-order decomposition is expected to a more accurate separation of the underlying contributions of EMG activity, ECG contamination, motion artifacts, and noise over time, QRS-sequence, and channels.

2.2.1. Identification of SVs related to disturbances

To identify ECG, artifacts, and noise components in the HO-SVD, we applied data-driven criteria in each mode (time within the RR interval - temporal, QRS-sequence, and channels - spatial).

Threshold selection on the temporal singular vector. In the temporal domain $U^{(1)}$, the goal was to identify components corresponding with ECG and artifact interference. Following the criterion introduced in SVD_{RI} [27], each singular vector was scored by a frequency ratio index (RI) [27], defined as the ratio of power below 30 Hz to power above 30 Hz, computed as

$$\mathrm{RI} = \frac{\mathrm{PSD}_{1-30\ \mathrm{Hz}}}{\mathrm{PSD}_{30-250\ \mathrm{Hz}}}, \tag{3}$$

where $\mathrm{PSD}_{1-30\ \mathrm{Hz}}$ and $\mathrm{PSD}_{30-250\ \mathrm{Hz}}$ are the spectral power below and above 30 Hz, respectively. Since ECG and motion artifacts typically have larger amplitudes than dEMG, and HO-SVD orders singular vectors from most to least influential, artifact-related components are expected to dominate the first singular vectors and are characterized by low-frequency content. Vectors with unusually large RI, outliers above the upper Tukey fence, were therefore flagged for removal. The threshold was calculated as

$$\mathrm{THR}_{RI} = \mathrm{Q}_3 + 1.5\ \mathrm{IQR}, \tag{4}$$

where Q_3 is a third quartile and IQR is an interquartile ratio. In the 2-D SVD case, this criterion alone was sufficient; however, extending

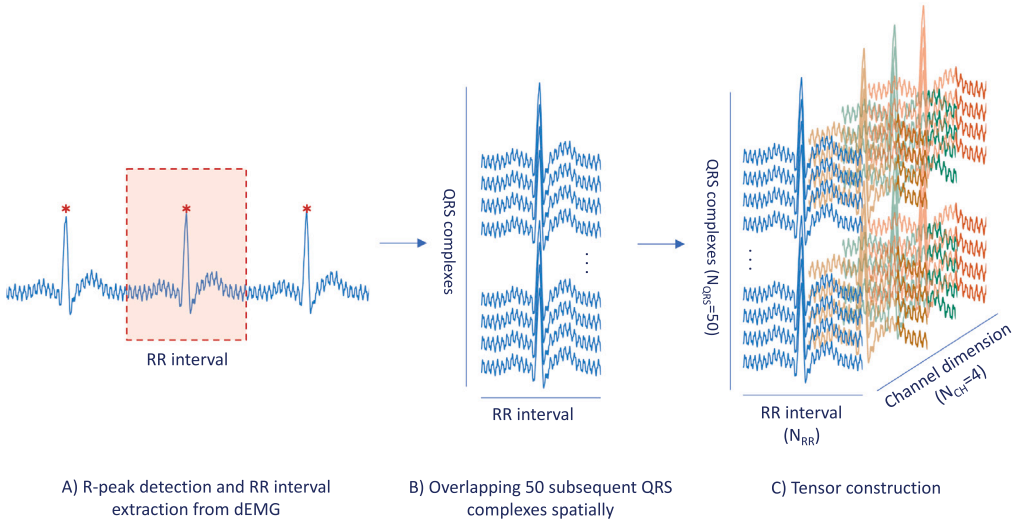


Fig. 1. Tensor construction. Panel A: R peaks are detected, and RR intervals are extracted from dEMG. Panel B: 50 Consecutive QRS complexes are overlapped spatially to form the QRS-sequence dimension ($N_{QRS} = 50$). Panel C: Data from multiple channels are incorporated to create a third-order tensor $T \in \mathbb{C}^{N_{RR} \times N_{QRS} \times N_{CH}}$, where N_{RR} denotes the number of samples in the RR interval, N_{QRS} the number of consecutive QRS complexes, and N_{CH} the number of channels.

the decomposition to a higher-order tensor distributes coherent ECG and artifact energy across multiple singular vectors within the temporal mode. As a result, artifact-related components often appear as a cluster rather than as a single isolated outlier. To capture this behavior, a contiguous cluster was formed starting at the first singular vector and extending through the outlier indices, but only as long as each subsequent outlier occurred within ≤4 singular-value positions of the previous one (based on the parameter sensitivity analysis; Section 3.1). This gap criterion prevents the inclusion of isolated distant outliers that are likely false positives. The highest index within this contiguous cluster was defined as the temporal cut-off m_{RR}.

Threshold selection on the QRS-sequence singular vector. In QRS-sequence mode $U^{(2)}$, the goal was to suppress the most coherent and repetitive components across QRS instances, which are captured in dominant singular vectors due to the consistent morphology of the ECG and motion artifacts. The coherence distribution in this mode was quantified using the singular values obtained from the second unfolded matrix T_2. These values were normalized by their total sum to obtain the percentage energy of each component representing its relative contribution to the overall coherence pattern across the QRS-sequence dimension. Components whose percentage energy exceeded the threshold were rejected (m_{QRS}), along with the immediately following index, to ensure the removal of adjacent high-energy contributions ($m_{QRS} + 1$). The threshold was defined as

$$\mathrm{THR}_{QRS} = \mathrm{Q}_3 + 0.75\ \mathrm{IQR}, \tag{5}$$

where a 0.75 IQR fence was selected to allow a more aggressive rejection in this dimension, acknowledging that ECG and motion artifacts, which are highly repeatable across QRS cycles, dominate the energy distribution and can mask the underlying dEMG activity if not sufficiently attenuated. The value of 0.75 IQR was determined based on the parameter sensitivity analysis presented in Section 3.1.

Threshold selection on the spatial singular vector. The spatial mode $U^{(3)}$ analysis aimed to identify and suppress components exhibiting high inter-channel coherence, as these represented ECG and artifacts rather than dEMG activity. The coherence distribution in this mode was estimated from the singular values obtained from the third unfolded tensor, T_3. To determine the amount of highly-coherent components, the 'knee' (change point) of the singular-value curve was identified. This knee point was defined as the position that minimizes the total fit error of two linear regressions fitted to the data segments on either side of each candidate break point. This procedure yielded the number of singular values to be removed from that dimension, denoted as $m_{\mathrm{CH}}+1$.

The suppression was then applied in the core tensor by zeroing all elements corresponding to the selected temporal, QRS-sequence and channel index ranges (1: m_{RR}, 1: m_{QRS} + 1, 1: m_{CH} + 1) before reconstruction. This combination targeted components that are (i) low-frequency within the RR window, (ii) dominant in terms of coherence as captured in the QRS-sequence mode, and (iii) highly correlated across channels in the spatial mode. Together, these characteristics highlight residual ECG, motion artifacts, and noise while the remaining components preserve dEMG-specific activity.

2.3. Performance assessment

The HO-SVD algorithm presented in this study was evaluated against $\mathrm{SVD}_{\mathrm{RI}}$, a method previously developed by our group and shown to outperform the original SVD approach as well as several established ECG-removal method [26,27]. As HO-SVD represents a further extension of this methodology, $\mathrm{SVD}_{\mathrm{RI}}$ was used as the reference. Additionally, a comparison of HO-SVD with two widely used state of the art methods, wavelet decomposition and adaptive template subtraction, was added in the Supplementary Material, in order to provide a broader assessment of the method.

Performance evaluation was carried out differently depending on the availability of the reference signal in the considered datasets. Regardless of the metrics chosen per dataset, in all cases non-parametric testing was used on the obtained results to assess whether the differences between the proposed HO-SVD method and the reference $\mathrm{SVD}_{\mathrm{RI}}$ algorithm were statistically significant. Normality was rejected using the Shapiro–Wilk test; therefore, paired comparisons were performed with the Wilcoxon signed-rank test. Results are reported as median values with interquartile ranges (IQRs).

2.3.1. Synthetic dEMG

The evaluation in the synthetic datasets (A,B,C) was based on the reference EMG signal. To quantify the reconstructed output ($\mathrm{SNR}_{\mathrm{REC}}$) was calculated as

$$\mathrm{SNR}_{REC} = 20\log_{10}\frac{\mathrm{std(EMG)}}{\mathrm{std}(\hat{\mathrm{EMG}} - \mathrm{EMG})}, \tag{6}$$

where EMG is the reference recordings, the $\hat{\mathrm{EMG}}$ is the reconstructed EMG after applying the HO-SVD and $\mathrm{SVD}_{\mathrm{RI}}$ methods. The residual noise

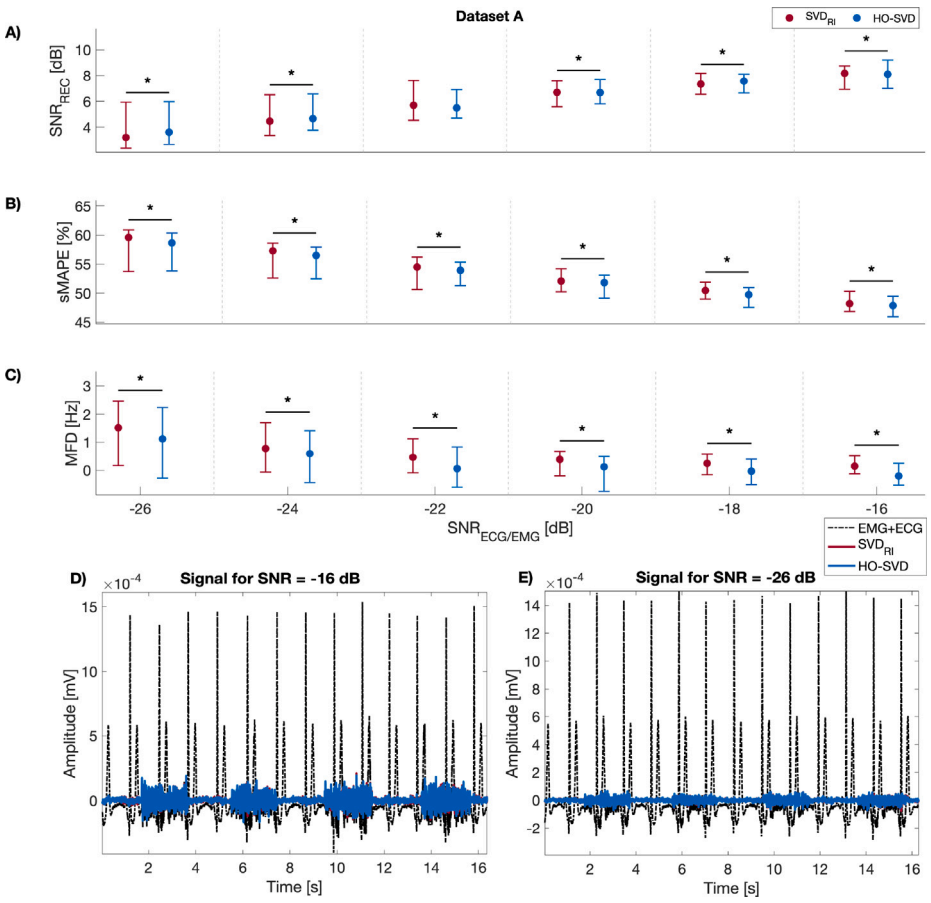


Fig. 2. Panel A–C: SNR_{REC}, sMAPE, and MFD for SVD_{RI} and HO-SVD in dataset A (range of SNR −26 to −16 dB). Panel D–E: an example of the reconstructed signal at −16 and −26 dB. In blue: performance and reconstruction for HO-SVD; red: performance and reconstruction for SVD_{RI}; black: data before reconstruction, * $p < 0.05$.

is estimated by subtracting the reference EMG from the reconstructed signal.

To determine amplitude errors, the symmetric mean absolute percentage error (sMAPE) was calculated. This metric reflects the precision of the muscular contraction level estimation [43]:

$$\text{sMAPE} = \frac{100\%}{N}\sum_{n=1}^{N}\frac{|\hat{\text{EMG}}_n - \text{EMG}_n|}{|\hat{\text{EMG}}_n| + |\text{EMG}_n|}, \tag{7}$$

where N is the number of samples. sMAPE was selected as a time-domain metric because it is bounded between 0%–100% and reduces the bias toward underestimated values. In the frequency domain, changes were evaluated using the mean frequency (MF), which is a well-established marker of myoelectric fatigue [44] and is given as

$$\text{MF} = \frac{\sum_{n=N_1}^{N_2} f_n S_n}{\sum_{n=N_1}^{N_2} S_n}, \tag{8}$$

where S_n is the spectral amplitude at a frequency bin f_n, derived from the short-time Fourier transform (STFT). The summation limits $[N_1, N_2]$ correspond to the band of 5–450 Hz. The MF values of the entire original and reconstructed EMG were compared using the mean frequency difference (MFD):

$$\text{MFD} = \text{MF}_{\text{EMG}} - \text{MF}_{\hat{\text{EMG}}}, \tag{9}$$

with MF_{EMG} and $\text{MF}_{\hat{\text{EMG}}}$ denoting the mean frequencies of the original and reconstructed signals, respectively. Together, sMAPE and MFD provided complementary perspectives on the time- and frequency-domain distortions introduced by ECG and artifact suppression.

2.3.2. Clinical dEMG

The results of the clinical data set were evaluated using three metrics.

The Spearman correlation coefficients (r) were computed between dEMG obtained from the two methods (HO-SVD, SVD_{RI}) and Pes envelope as a reference for golden-standard.

To determine the amount of ECG removed from the original signal, the ECG suppression index in the windows of 100 ms around each ECG peak was calculated as

$$\text{SI}_{\text{ECG}} = 20\log_{10}\frac{\text{mean}(\text{RMS}_{\text{EMG}})}{\text{mean}(\text{RMS}_{\hat{\text{EMG}}})}, \tag{10}$$

where RMS_{EMG} and $\text{RMS}_{\hat{\text{EMG}}}$ correspond to root mean square calculated within these windows in the original and reconstructed signals, respectively.

Lastly, the percentage of outliers in the reconstructed signals was calculated using Tukey's IQR method. Values lying outside the range $[Q_1 - 1.5\ \text{IQR}, Q_3 + 1.5\ \text{IQR}]$ were classified as outliers. The percentage of outliers was then determined as the ratio of the number of detected outliers to the total number of data points, multiplied by 100.

3. Results

3.1. Parameters sensitivity

The proposed HO-SVD framework employs three selection criteria across the temporal, QRS-sequence, and spatial modes. Of these, only the temporal gap criterion and the QRS-sequence Tukey-fence coefficient require to be determined. Therefore, a sensitivity analysis was

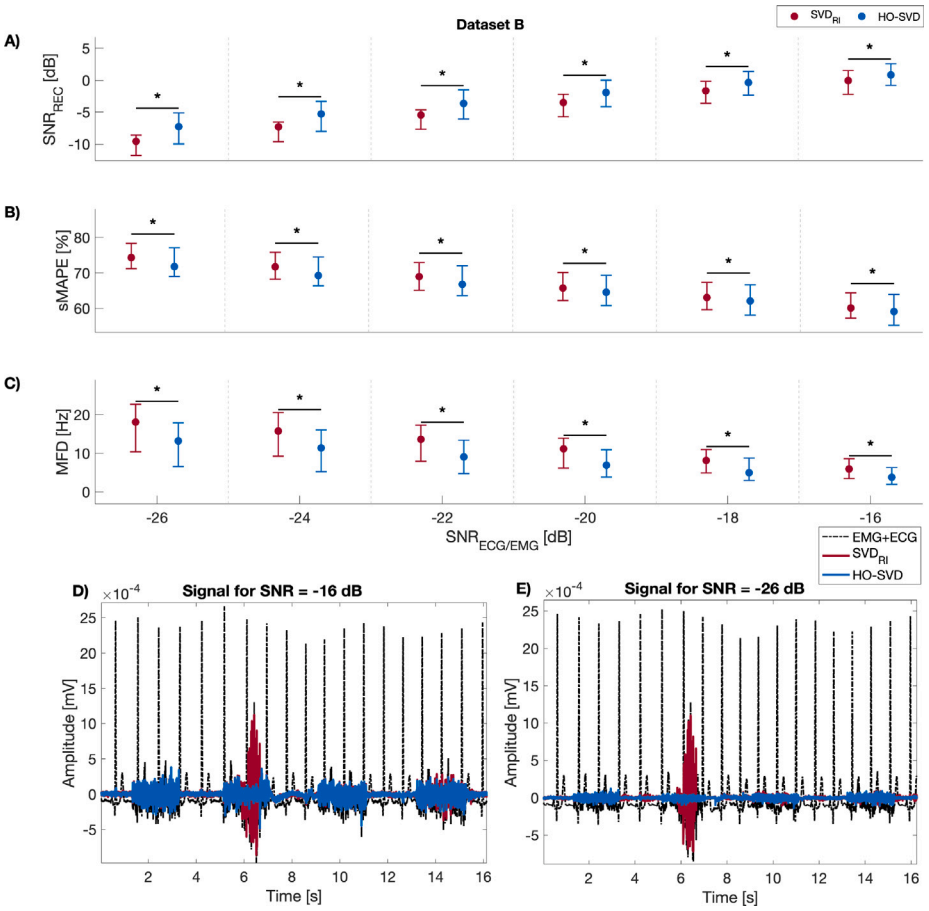


Fig. 3. Panel A–C: SNR_{REC}, sMAPE, and MFD for SVD_{RI} and HO-SVD in dataset B (range of SNR −26 to −16 dB). Panel D–E: an example of the reconstructed signal at −16 and −26 dB. In blue: performance and reconstruction for HO-SVD; red: performance and reconstruction for SVD_{RI}; black: data before reconstruction, * $p < 0.05$.

performed by varying the maximum allowed gap between consecutive outliers in the temporal mode (3, 4, and 5 singular-value positions) and the QRS-sequence coefficient (0.5, 0.75, and 1.0 IQR) across the three synthetic datasets. Across all tested values, the reconstruction quality remained stable, with maximum variations of 0.87 dB in SNR_{REC}, 0.013 in sMAPE, and 0.82 Hz in MFD. While individual parameter combinations occasionally produced slightly better results for specific datasets, no systematic performance degradation was observed. Based on these findings, the values of 0.75 IQR and a maximum gap of 4 singular-value positions were selected for the subsequent analyses, as they provided consistently good performance across all datasets.

3.2. Synthetic dEMG

Firstly, the algorithm was tested on the three datasets described in Section 2.1.1. The SNR results between −26 and −16 dB were computed for the HO-SVD algorithm and compared with the results obtained by the SVD ratio index algorithm (SVD_{RI}).

The results for dataset A are presented in Fig. 2. The SNR_{REC} (panel A) shows that both methods improve the SNR of the reconstructed signal, and the HO-SVD method performs significantly better ($p < 0.02$) with exception of input SNR −22 dB, where both methods obtain comparable results. The highest median gain between the input SNR and the SNR_{REC} reached 24 dB (151% increase), and it was registered for the HO-SVD algorithm with the starting SNR of −16 dB, achieving an output SNR of around 8 dB.

The signal reconstructed using the HO-SVD method presented significantly lower sMAPE (panel B) than SVD_{RI} for all input SNRs ($p <$ 0.0001). In the frequency domain, the median MFD (panel C) is always below 1.5 Hz and 1.2 Hz for SVD_{RI} and HO-SVD, respectively. The MFD values obtained for the HO-SVD method presented significantly better results than those for SVD_{RI} ($p < 0.0001$). Furthermore, two illustrative examples for SNR of −16 (panel D) and −26 dB (panel E) are reported. The reconstructed dEMG in panel D achieved a sMAPE of 47%, MFD of 0.3 Hz, and the SNR_{REC} of 9 dB for HO-SVD; and 49%, 0.4 Hz, and 8 dB for SVD_{RI}. The example shown in panel E for input SNR of −26 dB obtained 61% sMAPE and 3 dB SNR_{REC} for both methods, with MFD of 1.9 Hz and 2 Hz for HO-SVD and SVD_{RI}, respectively.

In dataset B, where motion artifacts affected the signal, the HO-SVD method performed significantly better in reconstructing EMG with higher SNR_{REC} ($p < 0.0001$) for the input SNR from −26 to −16 dB, as shown on Fig. 3. The new method achieved the highest median increase in the SNR of the reconstructed EMG equal to 19 dB (86% increase) for input SNR −22 dB, obtaining by that −2 dB in the output signal. The results obtained by sMAPE and MFD show a statistically better performance in favor of HO-SVD for all input SNRs. The signal examples in panels D and E show that the HO-SVD algorithm removed the ECG interference and motion artifacts for both −16 and −26 dB SNR. Differently, the SVD_{RI} algorithm successfully removed the ECG but not the motion artifacts. The example presented in panel D resulted in the HO-SVD method obtaining 46%, −0.1 Hz, 8 dB in the reconstructed dEMG for sMAPE, MFD, and SNR_{REC}, respectively. Following the SVD_{RI} achieved 49%, 7.5 Hz, and 3 dB. For the example presented in panel E, the results were 53%, 0.2 Hz and 7 dB for the HO-SVD algorithm, and 59%, 26 Hz, and −6 dB for SVD_{RI}.

The results for dataset C are presented in Fig. 4. Obtained SNR_{REF} was significantly higher for the HO-SVD method than for SVD_{RI} ($p <$

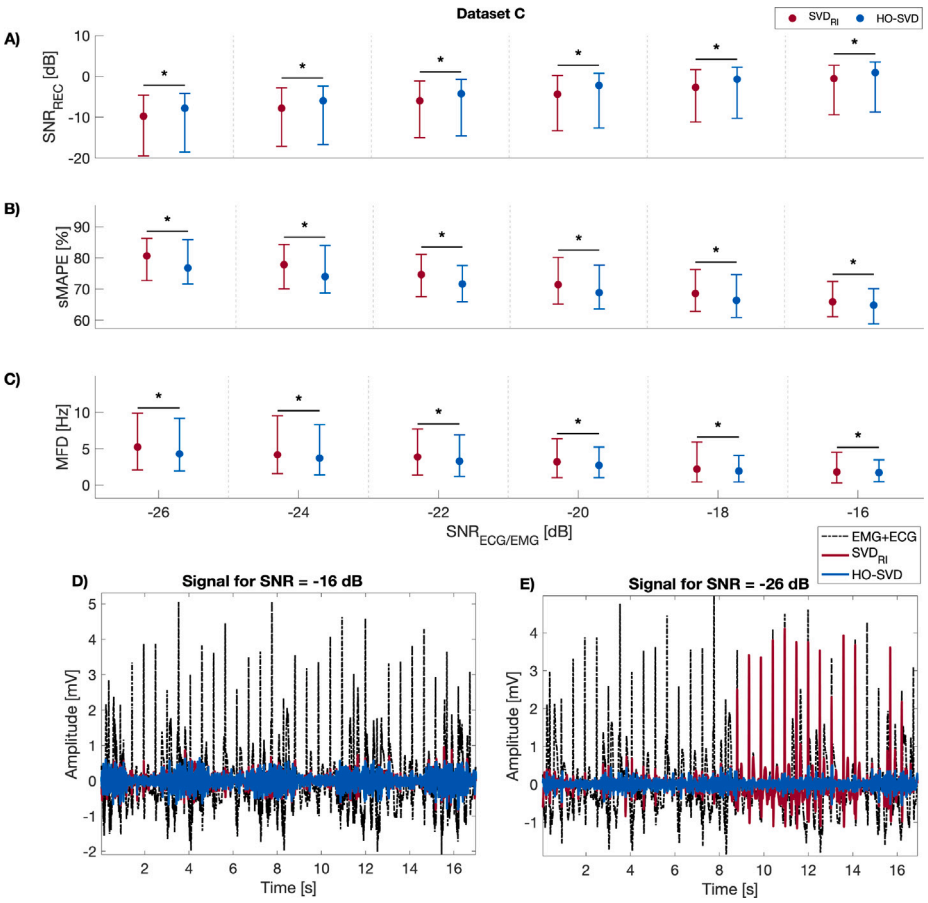


Fig. 4. Panel A–C: SNR_{REC}, sMAPE, and MFD for SVD_{RI} and HO-SVD in dataset C (range of SNR −26 to −16 dB). Panel D–E: an example of the reconstructed signal at −16 and −26 dB. In blue: performance and reconstruction for HO-SVD; red: performance and reconstruction for SVD_{RI}; black: data before reconstruction, * $p < 0.05$.

0.0001). The highest median increase in the SNR of the reconstructed EMG was achieved for the new method for the input SNR of −26 dB and was equal to 18 dB which resulted in 170% gain. The sMAPE and MDF metrics both showed significantly lower values for the HO-SVD algorithm compared to SVD_{RI}. Two illustrative examples are shown in Fig. 4 in panels D and E. Those resulted in 63% of sMAPE, 1.7 Hz of MFD, and 1 dB in SNR_{REC} for HO-SVD with input SNR of −16 dB. Correspondingly, the SVD_{RI} achieved 66%, 3.3 Hz, and −1 dB. Additionally, for panel E, the results were 76%, 88%; 9 Hz, 29 Hz; −8 dB, −22 dB; for HO-SVD and SVD_{RI} algorithms, respectively.

An additional supplementary comparison with two widely used and easily reproducible denoising methods, wavelet decomposition and adaptive template subtraction, was performed on the same synthetic datasets. The quantitative results are reported in the Supplementary Material.

3.3. Clinical dEMG

The results for 10 OSA patients are presented in Fig. 5. Panel A shows that the dEMG signal reconstructed using the HO-SVD method obtained significantly higher correlation with the Pes reference than the SVD_{RI} method ($p < 0.02$). The median correlation reached 0.56 for HO-SVD and 0.55 for SVD_{RI}. In panel B, HO-SVD showed significantly higher ECG suppression ($p < 0.0001$), with a median value of 15 dB. Lastly, the percentage of outliers in the reconstructed signal was significantly lower in the HO-SVD method than in the SVD_{RI} ($p < 0.0001$), with medians of 7% and 8%, respectively. Additionally, an example of the original signal and the two reconstructions is presented in panel D. This example clearly shows that the signal reconstructed using SVD_{RI} presents residual ECG activity, while the HO-SVD method provides EMG amplitude modulation without unwanted values.

4. Discussion

In this study, the HO-SVD algorithm was proposed to improve the removal of ECG interference, noise, and motion artifacts from the dEMG signal. The dimension-wise decomposition strategy provided a framework for identifying components corresponding directly to unwanted signals and removing them prior to dEMG reconstruction. The performance of the method was evaluated in the time and frequency domains using three synthetic datasets, for which the original EMG signal was available as reference. Additionally, the algorithm was tested on a clinical dataset to assess its performance on real-life data.

The results obtained for the dataset without artifacts or additional noise demonstrated a significant improvement in the reconstructed dEMG using the HO-SVD algorithm across all metrics, except at –22 dB input SNR, where the performance was comparable between methods. The HO-SVD method achieved a median improvement in the reconstructed SNR of 27 dB and an average median percentage gain of 131% relative to the input SNR values.

The datasets containing mild motion artifacts (B) and complex motion and muscle interference (C) showed that HO-SVD outperformed SVD_{RI} across all metrics and input SNR levels. As shown in panels D and E of Figs. 3 and 4, the dEMG reconstructed by SVD_{RI} still contained motion artifacts, ECG contamination, and signal distortion. In contrast, HO-SVD effectively removed all undesired components, allowing better dEMG reconstruction and preservation of its original amplitude modulation. The median improvement achieved by HO-SVD

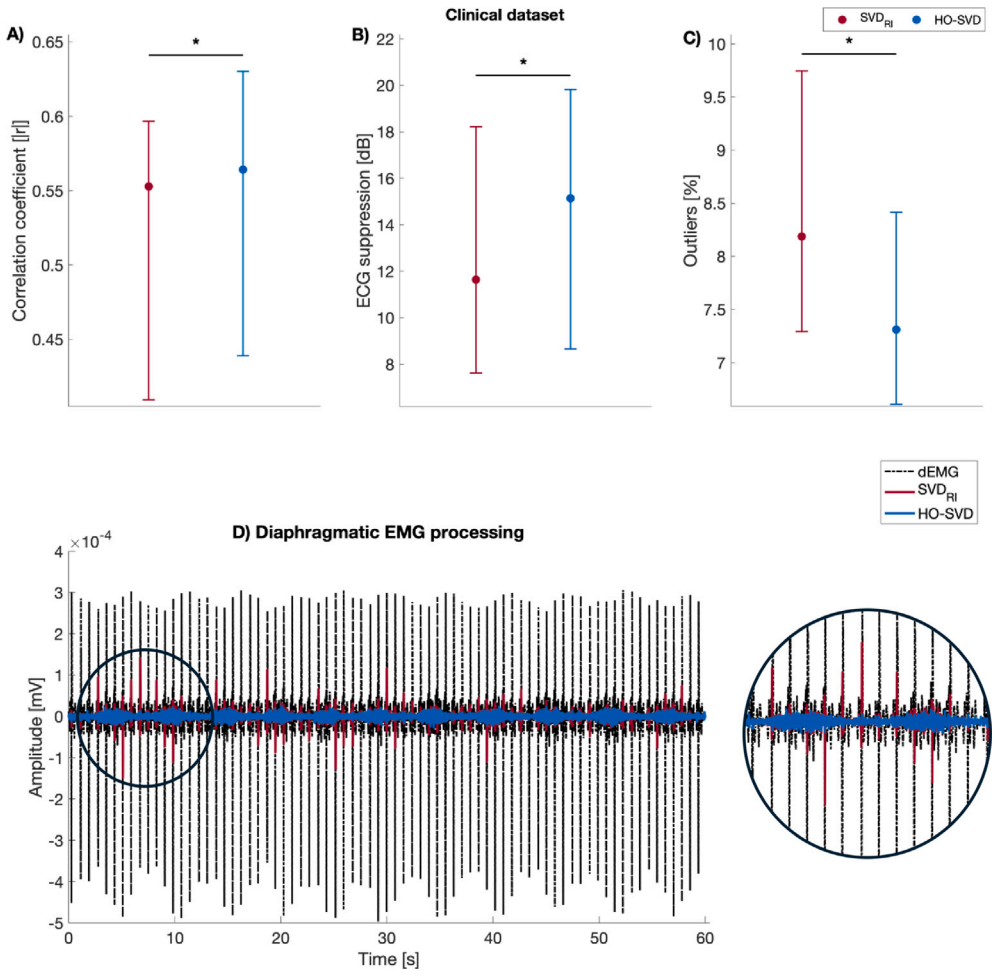


Fig. 5. Panel A, B, and C: Correlation coefficients with Pes reference, ECG suppression and Outliers for SVD_{RI}, and HO-SVD in the clinical dataset. Panel D: an example of the reconstructed signal. In blue: performance and reconstruction for HO-SVD; red: performance and reconstruction for SVD_{RI}; black: data before reconstruction, * $p < 0.05$.

for both datasets was 18 dB, corresponding to gains of 86% and 88% for datasets B and C, respectively.

While the synthetic datasets enabled controlled and systematic evaluation of artifact removal, they do not reflect the inherent physiological coupling between cardiac and diaphragmatic activity that occurs in real subjects. In these simulations, ECG and EMG signals originated from different individuals, meaning that the temporal alignment, propagation pathways, and amplitude relationships may not fully represent true cardiomuscular interactions. Furthermore, the synthetic EMG signals were derived from biceps and triceps recordings rather than diaphragm activity, and therefore may not fully reproduce the recruitment patterns and spectral characteristics of true dEMG. Additionally, the dimensionality mismatch between modalities - six EMG channels versus a single ECG recording used across all EMG channels - does not fully capture the spatial distribution and propagation of cardiac interference observed in real multichannel recordings. Moreover, motion artifacts in datasets B and C originated from ECG recordings containing motion-related contamination rather than from independently generated artifact sources. Consequently, the diversity of motion artifact scenarios encountered in clinical recordings may not be fully represented. Another limitation is that synthetic EMG signals do not include spontaneous respiratory variability, electrode-skin impedance changes, or cross-talk from neighboring muscles in the same form as in dEMG, all of which may influence decomposition performance in practice. These factors restrict the direct generalization of the synthetic-data results to clinical recordings. However, these constraints do not apply to the clinical dataset analysis presented below, where dEMG was obtained from OSA patients during full-night recordings.

The results obtained for the clinical dataset further confirmed the superiority of the proposed HO-SVD algorithm. The correlation with the reference gold standard (Pes) was significantly higher for the proposed method compared to SVD_{RI}. The moderate correlation observed between dEMG and Pes may partly reflect physiological differences between the two measurements, as Pes represents the cumulative pressure generated by all respiratory muscles. Although the diaphragm is the primary contributor to respiratory effort [45], accessory respiratory muscles may also be recruited, particularly during apneas, contributing to differences between dEMG and Pes. Moreover, in terms of ECG suppression and the outliers percentage in the reconstructed dEMG signal, SVD_{RI} remained significantly outperformed by HO-SVD. The illustrative example demonstrated that SVD_{RI} was unable to completely remove ECG contamination, resulting in residual peaks in the reconstructed signal. This residual activity may be erroneously attributed to muscle contractions, thus distorting the true amplitude modulation of dEMG. In contrast, the HO-SVD approach effectively eliminated ECG contamination while preserving the original modulation of the dEMG signal.

Considering all these findings, the proposed HO-SVD algorithm clearly outperforms the SVD_{RI} approach when the signal is affected by ECG interference, but also when motion artifacts, or additional noise are present, reconstructing the dEMG signal with significantly higher quality across all tested conditions. The results confirm that the

incorporation of a third spatial dimension, paired with the possibility to finely adjust the choice of the thresholds through the temporal, QRS-sequence, and spatial dimensions, leads to a notable improvement in the ECG, motion artifacts, and noise removal, preserving a higher-quality dEMG reconstruction.

Future work could extend this analysis by including larger clinical cohorts and assessing performance against additional physiological references, such as esophageal pressure. Additionally, a direct quantitative comparison with recently proposed dEMG denoising methods, such as deep learning-based approaches [46], represents an important direction for future evaluation. Finally, the proposed HO-SVD framework could be explored for other biomedical applications where artifact suppression is crucial, including ECG or multi-channel EEG, where improved signal quality would enhance both interpretation and analysis.

In summary, this study introduced an HO-SVD framework for the suppression of ECG interference, as well as low and high-intensity motion artifacts in respiratory dEMG. By exploiting multidimensional decomposition across signal domains, the method enabled selective removal of unwanted components while preserving physiological signal integrity. Evaluation on synthetic and clinical datasets demonstrated that HO-SVD consistently improved reconstruction quality and correlation with the reference signal compared to the conventional SVD approach. These results confirm the algorithm's potential as a robust and flexible tool for enhancing dEMG signal quality in respiratory research. Future work will focus on optimizing parameter selection, validating the method in larger clinical cohorts, and exploring its applicability to other biomedical modalities such as ECG and multi-channel EEG.

CRediT authorship contribution statement

Gabriela Grońska: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Elisabetta Peri:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Xi Long:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Sebastiaan Overeem:** Writing – review & editing, Data curation, Conceptualization. **Johannes P. van Dijk:** Writing – review & editing, Data curation, Conceptualization. **Massimo Mischi:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Ethics declaration

The recordings used in this study were obtained under a protocol approved by the Ethics Committee of the Máxima Medical Center (Veldhoven, the Netherlands; Approval No. N16.074). The study was conducted in accordance with the Declaration of Helsinki and applicable institutional guidelines. Written informed consent to participate in the study and to publish the article was obtained from all participants. Participants' privacy rights have been observed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cmpb.2026.109559.

References

[1] N. Bockelmann, et al., Deep learning for prediction of diaphragm activity from the surface electromyogram, Curr. Dir. Biomed. Eng. 5 (1) (2019) 17–20.

[2] N. Vandenbussche, S. Overeem, J. van Dijk, P. Simons, D. Pevernagie, Assessment of respiratory effort during sleep: Esophageal pressure versus noninvasive monitoring techniques, Sleep Med. Rev. 24 (2015) 28–36.

[3] E. Clancy, E. Morin, R. Merletti, Sampling, noise-reduction and amplitude estimation issues in surface electromyography, J. Electromyogr. Kinesiol. 12 (2002) 1–16, http://dx.doi.org/10.1016/S1050-6411(01)00033-5.

[4] J. Drake, J. Callaghan, Elimination of electrocardiogram contamination from electromyogram signals: An evaluation of currently used removal techniques, J. Electromyogr. Kinesiol. 16 (2006) 175–187, http://dx.doi.org/10.1016/j.jelekin.2005.07.003.

[5] A. Bartolo, C. Roberts, R. Dzwonczyk, E. Goldman, Analysis of diaphragm EMG signals: Comparison of gating vs. subtraction for removal of ECG contamination, J. Appl. Physiol. 80 (1996) 1898–1902, http://dx.doi.org/10.1152/jappl.1996.80.6.1898.

[6] A. Jonkman, et al., Analysis and applications of respiratory surface EMG: report of a round table meeting, Crit. Care 28 (1) (2024) 2, http://dx.doi.org/10.1186/s13054-023-04779-x.

[7] I. Christov, I. Daskalov, Filtering of electromyogram artifacts from the electrocardiogram, Med. Eng. Phys. 21 (1999) 731–736.

[8] M. Redfern, R. Hughes, D. Chaffin, High-pass filtering to remove electrocardiographic interference from torso EMG recordings, Clin. Biomech. 8 (1993) 44–48, http://dx.doi.org/10.1016/S0268-0033(05)80009-9.

[9] L. Xu, E. Peri, R. Vullings, C. Rabotti, J. Van Dijk, M. Mischi, Comparative review of the algorithms for removal of electrocardiographic interference from trunk electromyography, Sensors 20 (17) (2020) 4890, http://dx.doi.org/10.3390/s20174890.

[10] S. Levine, J. Gillen, P. Weiser, M. Gillen, E. Kwatny, Description and validation of an ECG removal procedure for EMGdi power spectrum analysis, J. Appl. Physiol. 60 (1986) 1073–1081, http://dx.doi.org/10.1152/jappl.1986.60.3.1073.

[11] M. Nitzken, N. Bajaj, S. Aslan, G. Gimel'farb, A. El-Baz, A. Ovechkin, Local wavelet-based filtering of electromyographic signals to eliminate the electrocardiographic-induced artifacts in patients with spinal cord injury, J. Biomed. Sci. Eng. 6 (7A2) (2013) 1–13, http://dx.doi.org/10.4236/jbise.2013.67A2001.

[12] W. Lu, D. Gong, X. Xue, L. Gao, Improved multi-layer wavelet transform and blind source separation based ECG artifacts removal algorithm from the sEMG signal: in the case of upper limbs, Front. Bioeng. Biotechnol. 12 (2024) 1367929, http://dx.doi.org/10.3389/fbioe.2024.1367929.

[13] J. Cardoso, Blind signal separation: Statistical principles, Proc. IEEE 86 (10) (1998) 2009–2025.

[14] J. Mak, Y. Hu, K. Luk, An automated ECG-artifact removal method for trunk muscle surface EMG recordings, Med. Eng. Phys. 32 (2010) 840–848, http://dx.doi.org/10.1016/j.medengphy.2010.05.007.

[15] N. Willigenburg, A. Daffertshofer, I. Kingma, J. van Dieën, Removing ECG contamination from EMG recordings: A comparison of ICA-based and other filtering procedures, J. Electromyogr. Kinesiol. 22 (3) (2012) 485–493, http://dx.doi.org/10.1016/j.jelekin.2012.01.001.

[16] M. Al Harrach, S. Boudaoud, M. Hassan, F.S. Ayachi, D. Gamet, J.F. Grosset, et al., Denoising of HD-sEMG signals using canonical correlation analysis., Med. Biol. Eng. Comput. 55 (3) (2017) 375–388, http://dx.doi.org/10.1007/s11517-016-1521-x.

[17] D. Anand, V. Bhateja, A. Srivastava, D.K. Tiwari, An approach for the preprocessing of EMG signals using canonical correlation analysis, in: S.C. Satapathy, V. Bhateja, S. Das (Eds.), Smart Computing and Informatics, in: Smart Innovation, Systems and Technologies, 78, Springer, Singapore, 2018, pp. 201–208, http://dx.doi.org/10.1007/978-981-10-5547-8_21.

[18] K. Wang, X. Chen, L. Wu, X. Zhang, Z. Wang, High-Density surface EMG denoising using independent vector analysis, IEEE Trans. Neural Syst. Rehabil. Eng. 28 (6) (2020) 1271–1281, http://dx.doi.org/10.1109/TNSRE.2020.2987709.

[19] J.D. Costa Júnior, D.D. Ferreira, J. Nadal, A.M.F.L. Miranda de Sá, Reducing electrocardiographic artifacts from electromyogram signals with independent component analysis, in: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), 2010, pp. 4598–4601.

[20] Y. Hu, X.H. Li, X.B. Xie, L.Y. Pang, Y. Cao, K.D.K. Luk, Applying independent component analysis on ECG cancellation technique for the surface recording of trunk electromyography, in: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), 2005, pp. 3647–3649.

[21] S.R. Alty, W.D.C. Man, J. Moxham, K.C. Lee, Denoising of diaphragmatic electromyogram signals for respiratory control and diagnostic purposes, in: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), 2008, pp. 5560–5563, http://dx.doi.org/10.1109/IEMBS.2008.4650474.

[22] M. Chen, X. Zhang, X. Chen, M. Zhu, G. Li, P. Zhou, FastICA peel-off for ECG interference removal from surface EMG, Biomed. Eng. Online 15 (2016) 65, http://dx.doi.org/10.1186/s12938-016-0196-8.

[23] P. Kanjilal, S. Palit, G. Saha, Fetal ECG extraction from single-channel maternal ECG using singular value decomposition, IEEE Trans. Biomed. Eng. 44 (1997) 51–59.
[24] M.A. Ouali, K. Chafaa, SVD-based method for ECG denoising, in: Proc. Int. Conf. Comput. Appl. Technol. (ICCAT), 2013, pp. 1–4.
[25] M. Bandarabadi, M. Karami-Mollaei, A. Afzalian, J. Ghasemi, ECG denoising using singular value decomposition, Aust. J. Basic Appl. Sci. 4 (2010) 2109–2113.
[26] E. Peri, et al., Singular value decomposition for removal of cardiac interference from trunk electromyogram, Sensors 21 (2) (2021) 573, http://dx.doi.org/10.3390/s21020573.
[27] G. Grońska, E. Peri, X. Long, J. van Dijk, M. Mischi, Removal of electrocardiographic interference and artifacts from diaphragm electromyography, in: 2023 IEEE Int. Symp. Med. Meas. Appl. (MeMeA), IEEE, 2023, pp. 1–6, http://dx.doi.org/10.1109/MeMeA57477.2023.10171861.
[28] F. Sammali, et al., Dedicated ultrasound speckle tracking for quantitative analysis of uterine motion outside pregnancy, IEEE Trans. Ultrason. 66 (3) (2019) 581–590.
[29] R. Wildeboer, et al., Blind source separation for clutter and noise suppression in ultrasound imaging: Review for different applications, IEEE Trans. Ultrason. 67 (8) (2020) 1497–1512.
[30] F. Sammali, et al., Multi-Modal uterine-activity measurements for prediction of embryo implantation by machine learning, IEEE Access 9 (2021) 47096–47111.
[31] K. Ozgun, B. Byram, Multidimensional clutter filtering of aperture domain data for improved blood flow sensitivity, IEEE Trans. Ultrason. Ferroelectr. Freq. Control 68 (8) (2021) 2645–2656.
[32] M. Kim, Y. Zhu, J. Hedhli, L. Dobrucki, M. Insana, Multidimensional clutter filter optimization for ultrasonic perfusion imaging, IEEE Trans. Ultrason. Ferroelectr. Freq. 65 (11) (2018) 2020–2029.
[33] Y. Huang, X. Chen, E. Badescu, M. Kuenen, O. Bonnefous, M. Mischi, Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion, Ultrasonics 140 (2024) 107307, http://dx.doi.org/10.1016/j.ultras.2024.107307.
[34] L. Yang, Y. Wang, New method for diffusion-weighted images denoising based on patch-matching with higher-order singular value decomposition, J. X-Ray Sci. Technol. 33 (3) (2025) 526–539, http://dx.doi.org/10.1177/08953996241313321.
[35] L. Thanh, N. Dao, N. Dung, N. Trung, K. Abed-Meraim, Multi-channel EEG epileptic spike detection by a new method of tensor decomposition, J. Neural Eng. 17 (1) (2020) 016023, http://dx.doi.org/10.1088/1741-2552/ab5247.
[36] M. Cretikos, et al., Respiratory rate: the neglected vital sign, Med. J. Aust. 188 (11) (2008) 657–659.
[37] M. van Gilst, et al., Protocol of the SOMNIA project: An observational study to create a neurophysiological database for advanced clinical sleep monitoring, BMJ Open 9 (11) (2019) e030996, http://dx.doi.org/10.1136/bmjopen-2019-030996.
[38] R. Merletti, P. Parker (Eds.), Electromyography: Physiology, engineering, and non-invasive applications, John Wiley & Sons, Hoboken, 2004.
[39] L. Sörnmo, P. Laguna, Electrocardiogram ECG signal processing, in: M. Akay (Ed.), Wiley Encyclopedia of Biomedical Engineering, 2, John Wiley & Sons, 2006, pp. 1298–1313, http://dx.doi.org/10.1002/9780471740360.ebs1482.
[40] L. Tucker, Some mathematical notes on three-mode factor analysis, Psychometrika 31 (1966) 279–311.
[41] N. Vervliet, O. Debals, L. Sorber, M. van Barel, L. De Lathauwer, Tensorlab 3.0, 3.0, 2016, URL ttps://www.tensorlab.net.
[42] I. Silva, J. Behar, R. Sameni, T. Zhu, J. Oster, G. Clifford, G. Moody, Noninvasive fetal ECG: The PhysioNet/Computing in Cardiology Challenge 2013, in: Proc. Comput. Cardiol., 40, 2013, pp. 149–152.
[43] E. Clancy, N. Hogan, Multiple site electromyograph amplitude estimation, IEEE Trans. Biomed. Eng. 42 (2) (1995) 203–211.
[44] L. Xu, C. Rabotti, M. Mischi, Analysis of vibration exercise at varying frequencies by different fatigue estimators, IEEE Trans. Neural Syst. Rehabil. Eng. 24 (12) (2016) 1284–1293, http://dx.doi.org/10.1109/TNSRE.2015.2493358.
[45] A. De Troyer, A. Boriek, Mechanics of the respiratory muscles, Compr. Physiol. 1 (3) (2011) 1273–1300.
[46] M.G. Vorborg, J. de Araujo, S. Puthusserypady, S.S. Rasmussen, U-Net based decomposition of respiratory surface electromyography for removal of cardiac artifacts, in: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC), 2025.